

Mini-Assessment

Designing a Multi-Stage AI Prompt Workflow

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Assignment: Mini Assessment — Prompt Workflow Design

1. Assessment Goal

Build a repeatable, six-stage prompt workflow that can be run entirely inside an AI chat interface. The workflow splits one broad deliverable into separate AI-driven stages, where the output of each stage becomes the direct input to the next, so quality is built up gradually instead of requested in a single ask.

2. Workflow Overview

Step	Purpose	Main Output
1	Frame the problem	Problem statement + scope
2	Design the approach	Action roadmap
3	Produce the first output	Working draft
4	Stress-test the output	Gap list + risk notes
5	Strengthen the output	Polished version
6	Pre-submission audit	Verified, ready-to-submit output

Step 1 — Frame the Problem

What happens

Before anything is produced, the AI is asked to pin down exactly what is being solved: the objective, the boundaries of the task, who it's for, and what "done" looks like. No solution is generated at this stage — only a clear frame for the work ahead.

Prompt to use in the AI UI

You are the problem-framing stage of a multi-step AI workflow. Read the request below and produce: 1. The core objective, stated in one sentence 2. What is explicitly in scope and out of scope 3. The intended audience or end user 4. Any constraints, rules, or format requirements 5. A short definition of what a successful result looks like Request: [PASTE THE REQUEST HERE] Do not attempt a solution yet. Only produce the problem frame for the next stage.

Expected output

A short problem-framing brief that becomes the fixed reference point for every later stage.

Step 2 — Design the Approach

What happens

With the problem framed, the AI now maps out how it will get there: the sequence of work, the information or resources each part needs, and where things commonly go wrong. This is planning only — still no finished content.

Prompt to use in the AI UI

Using the problem frame below, design a step-by-step approach for tackling it.
Include:

- The main pieces of work, in the order they should happen
- What information or inputs each piece needs
- The expected result of each piece of work
- Common pitfalls to watch for at this stage

Problem frame:

[PASTE STEP 1 OUTPUT HERE]

Keep the plan concrete and directly usable – no generic advice.

Expected output

A practical roadmap that the next stage will follow to produce a first output.

Step 3 — Produce the First Output

What happens

The AI now generates a complete first version of the actual deliverable, strictly following the roadmap and the original constraints from Step 1. This is a real, usable draft — not a summary of intent.

Prompt to use in the AI UI

Produce a complete first version of the deliverable described below.
Follow the problem frame and the roadmap exactly – do not skip requirements.
Use clear structure and appropriate formatting for the intended audience.
Produce the actual deliverable itself, not a description of what you would do.

Problem frame:

[PASTE STEP 1 OUTPUT]

Roadmap:

[PASTE STEP 2 OUTPUT]

Expected output

A working first draft, complete enough to be evaluated on its own merits.

Step 4 — Stress-Test the Output

What happens

Rather than trusting the first draft, a separate pass asks the AI to attack it as a skeptical reviewer — checking it against the original frame, hunting for gaps, weak logic, and anything that would fail if it were graded or shipped.

Prompt to use in the AI UI

Act as a skeptical, detail-focused reviewer.
Compare the draft below against the original problem frame and roadmap.
Check for:

- Requirements that were missed or only partly addressed
- Logical gaps or unsupported claims
- Unclear or inconsistent sections
- Structural or formatting issues
- Anything a strict evaluator would flag

Return:

- A. Strengths of the draft
- B. Specific weaknesses, with examples
- C. Missing requirements
- D. Concrete fixes for each issue
- E. An overall readiness score out of 10

Do not rewrite the draft – only diagnose it.

Problem frame: [PASTE STEP 1 OUTPUT]
Draft: [PASTE STEP 3 OUTPUT]

Expected output

A diagnostic review report with a specific, actionable fix list.

Step 5 — Strengthen the Output

What happens

The review feedback is fed back in alongside the original draft, and the AI produces an improved version — applying valid fixes, keeping what already worked, and not inventing new requirements that were never asked for.

Prompt to use in the AI UI

Produce an improved version of the deliverable using everything below.
Apply every valid fix from the review.
Keep any part of the draft that already meets the requirements – don't rewrite unnecessarily.
Do not introduce requirements that were not in the original problem frame.

Problem frame: [PASTE STEP 1 OUTPUT]
Draft: [PASTE STEP 3 OUTPUT]
Review report: [PASTE STEP 4 OUTPUT]

Return only the improved deliverable, ready for final checking.

Expected output

A strengthened, near-final version that has already absorbed the critique.

Step 6 — Pre-Submission Audit

What happens

Before treating the work as finished, one last pass verifies — requirement by requirement — that the improved version actually satisfies the original frame, rather than assuming the strengthening stage got everything right.

Prompt to use in the AI UI

Perform a final audit of the deliverable below before it is submitted.
Go through the original requirements one by one and mark each:
- PASS – fully satisfied
- FIX – missing, incomplete, or incorrect
Also check overall clarity, structure, consistency, and formatting.
For every FIX, state exactly what needs to change.

Problem frame: [PASTE STEP 1 OUTPUT]
Final deliverable: [PASTE STEP 5 OUTPUT]

Expected output

A requirement-by-requirement checklist. If everything is marked PASS, the deliverable is ready to submit.

3. How to Run the Workflow in the UI

- Open the AI chat interface and start with Step 1 as its own message.
- Copy each step's output and paste it into the next step's prompt exactly where indicated.
- Keep the original request visible or saved somewhere so requirements are never lost between stages.
- Never merge Step 4 (review) into Step 3 (drafting) — separating them is what makes the critique honest.
- Only treat the result as final once Step 6 returns all PASS marks.

4. Workflow Logic

Request → Framing → Roadmap → First Output → Stress-Test → Strengthened Output → Audit

This structure keeps reasoning, generation, and quality control as separate stages instead of one blended request. Each stage has exactly one job and one defined output, which is what makes the workflow demonstrable step-by-step in a plain AI interface with no additional tooling.

5. Assessment Submission Checklist

- The workflow contains multiple clearly defined AI stages.
- Every stage has its own specific prompt and single purpose.
- Each stage's output is explicitly reused as input to the next stage.
- The original request/requirements stay visible throughout the process.
- A dedicated critique/stress-test stage is included, separate from drafting.
- A dedicated final audit stage is included before submission.
- No code or software build is required — everything runs through prompts.
- The workflow can be demonstrated live in any AI chat interface.